Dual-Spin Fixed-Canard Precision Guidance Kit for 155 mm Artillery: Dynamics, Guidance, Navigation, Control, and Fuze Logic

Abstract

This paper presents a reduced-order six-degree-of-freedom (6DOF) dual-spin model of an M1156-class precision guidance kit (PGK) for 155 mm artillery. A high-spin aft body provides gyroscopic stability; a de-spun nose carries fixed canards under free-spin / hold control (no continuous fin servo). Guidance uses zero-effort miss (ZEM) with a simplified impact predictor. Navigation is GPS-dominant with a six-state constant-velocity Kalman filter; nose-roll sensing closes the hold loop. The multi-mode fuze supports impact, time, and proximity functions with setback safe–arm and early detonation inhibit on gross predicted miss. For a representative long-range case, open-loop range is about 15.7 km and open-loop miss 122 m to an offset target; guided miss is about 13 m under truth and GPS navigation, within a < 30 m single-shot objective band.

Keywords: dual-spin projectile; fixed canard; PGK; ZEM; GPS; free-spin/hold; multi-mode fuze

1. System Architecture

1.1 Dual-spin fixed-canard concept

The aft body remains at high spin (order 100–180 rps at launch, decaying in flight) for stability. The nose section is mechanically de-spun so fixed canards can hold a commanded roll orientation and produce a quasi-steady lateral force. Relative spin may drive an alternator for power. Control is free-spin (mean canard lift cancels) or hold (brake/phase-hold; steady side force). Force direction uses physical nose attitude φ_nose, not command θ_c alone.

1.2 GNC block diagram

Figure 0 summarises the closed loop: GPS measurements feed a constant-velocity Kalman filter; the filter state drives a ZEM impact predictor; the mode manager selects guide or free-spin; hold tracks θ_c using measured nose roll; the fuze runs in parallel with independent inhibit logic.

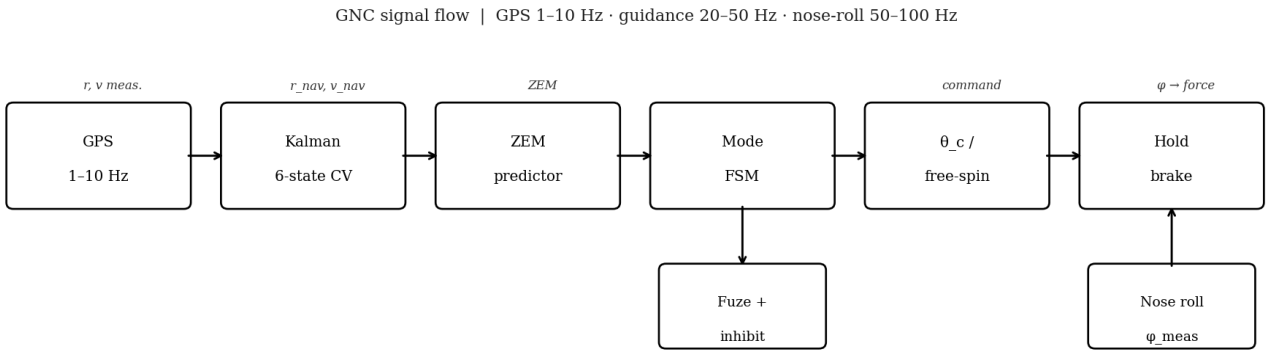


Figure 0. GNC and fuze signal flow. Rates: GPS 1–10 Hz; guidance/modes 20–50 Hz; nose-roll sense 50–100 Hz class.

1.3 Sensors

Sensor	Role	Notes
GPS L1	Primary nav	1–10 Hz, ~50 ms latency
Nose roll	Hold feedback	Required for HOLD mode
Setback	Safe–arm / launch	Fuze + mode FSM
Impact / timer / prox	Fuze modes	PD, time, HOB
IMU (optional)	Coast only	Not baseline GNC

Table 1. Baseline sensor selection.

1.4 Flight software rates and mode FSM

Software is organised by rate groups. GPS runs at 1–10 Hz with modelled latency. The navigation filter predicts every control tick and updates on GPS. Guidance and mode logic run at the control interval (about 20–50 Hz). Nose-roll sensing and hold torque operate at a higher rate (physics step or 50–100 Hz class). The fuze is polled at a high rate and reacts to events (setback, height, impact).

Mode	Condition (summary)	Canards / guidance
SAFE	Pre-launch / before setback	Inactive
COAST	Post-launch, before arm time	FREE-SPIN
GUIDE	Armed, ZEM above deadband, nav OK	HOLD toward θ_c
FREE_SPIN	ZEM inside deadband or coast	Spin; mean force ~0
FAILSAFE	Gross early ZEM or GPS stale	FREE-SPIN; ballistic
TERMINAL	Fuze fire or impact	End of control

Table 2. Mode manager (FSM) summary.

Fail-safe: (i) if early predicted miss exceeds a large threshold (order hundreds of metres), guidance is inhibited and the shell remains free-spin ballistic; (ii) if GPS is stale beyond a watchdog (order 2 s), the mode manager forces free-spin. Fuze detonation inhibit on ~150 m early miss is separate from guidance fail-safe and can block the explosive train even if impact occurs.

2. Dynamics and Aerodynamics

State: position, velocity, quaternion attitude, body rates, nose roll angle/rate. Rigid-body equations include gravity, optional Coriolis, Mach-dependent body aero, and canard force/moment. RK4 integration uses $\Delta t \approx 2.5$ ms.

Canard lateral coefficient: $C_Y = C_{N0} \cdot f_\alpha(\alpha) \cdot f_M(M) \cdot f_{spin}(\omega_{nose}, mode)$, attenuated in free-spin and while spinning down. Force is capped by q and a_max. Nominal freeze: phase = 0°, arm ≈ 7 s, CN ≈ 0.027, a_max ≈ 1.05 m/s².

3. Guidance, Navigation and Control

3.1 Navigation

GPS discrete fixes update a six-state constant-velocity Kalman filter (position and velocity). Guidance uses filter outputs in GPS mode. Nose-roll measurements close the hold loop.

3.2 Guidance law

A simplified ballistic predictor (flat drag) yields predicted impact from nav state. ZEM is horizontal miss to target. If ZEM exceeds the deadband, $\theta_c = \text{atan2}(ZEM_y, ZEM_x) + \varphi_{phase}$; else free-spin. Phase was frozen at 0° after closed-loop tests.

3.3 Flight control

Outer loop: ZEM → θ_c or free-spin. Inner loop: hold brake drives $\omega_{nose} \rightarrow 0$ and $\varphi_{nose} \rightarrow \theta_c$. No continuous canard incidence servo.

4. Multi-Mode Electronic Fuze

IMPACT (ground contact after arming); TIME (programmed time from setback); PROXIMITY (height ≤ HOB, impact fallback). Safe–arm: setback plus short separation delay. Detonation inhibit if early predicted miss exceeds ~150 m.

5. Simulation Results

5.1 Summary metrics

Quantity	Value
Open-loop range	15 714 m
Open-loop drift (Y)	265 m
Open-loop miss (offset target)	122.1 m
Guided miss (truth nav)	13.3 m
Guided miss (GPS)	12.6 m
Guidance phase / arm time	0° / 7 s
Approx. CN / a_{max}	0.027 / 1.05 m/s²
PGK objective / threshold	< 30 m / < 50 m

Table 3. Nominal closed-loop scenario (target offset ~100 m range, ~70 m cross-range).

5.2 Control budget and hold quality

Authority realism is reported via canard-specific effort. For a typical guided run, integrated lateral ΔV while holding is on the order of tens of metres per second, mean canard acceleration remains below 1 m/s², and maximum angle of attack stays a few degrees.

Metric	Typical guided run (order of)
Lateral ΔV while HOLD	~15–25 m/s
Mean |a_can| while HOLD	~0.3–0.5 m/s²
Max |a_can|	< a_max (1.05 m/s²)
Max |AoA|	~3–4°
Time to de-spin after arm	~7–8 s flight time (~1 s after arm)
HOLD fraction after arm	high (≳ 90%)

Table 4. Control budget and hold-quality orders of magnitude.

Guided miss falls from 122 m open-loop to about 13 m (truth and GPS), below the 30 m single-shot objective. Formal CEP_{50} still requires Monte Carlo with $N \geq 30$.

5.3 Trajectories and dual-spin behaviour

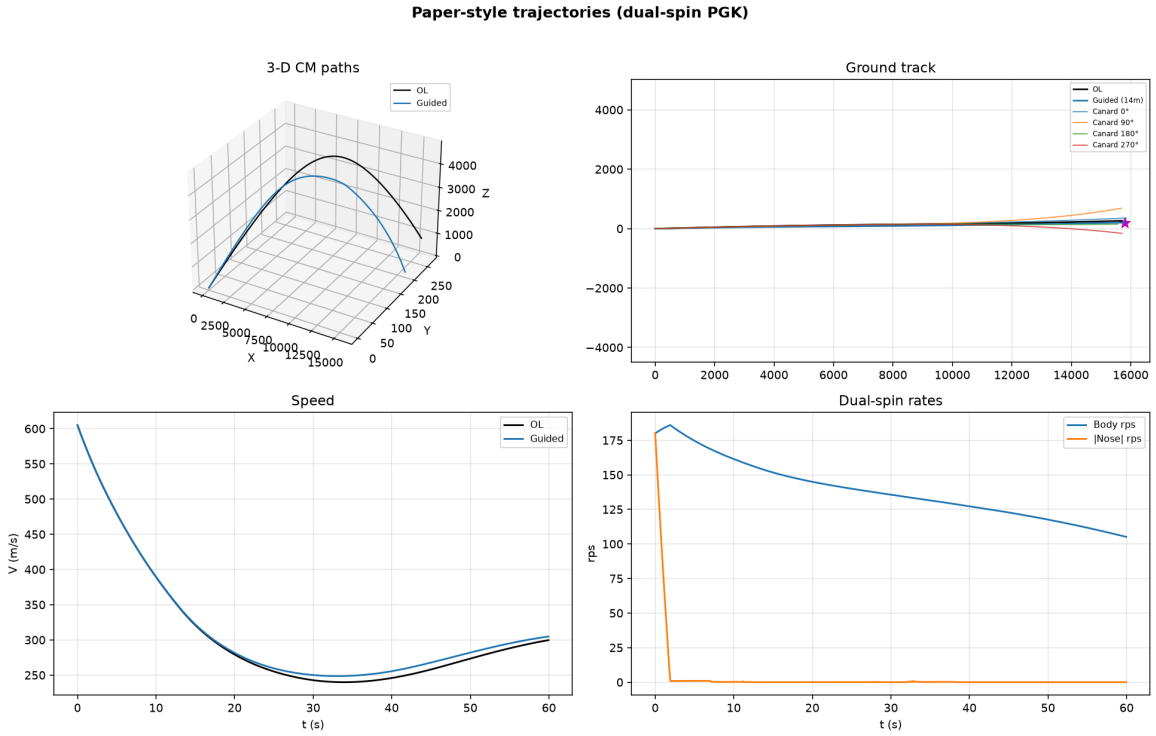


Figure 1. Paper-style trajectories: 3-D paths, ground track, speed, dual-spin rates.

The centre-of-mass path is a smooth ballistic arc. Body spin decays from ~180 rps; the nose de-spins rapidly under hold.

Figure 2. Calibration workflow: ground track, miss vs 30/50 m bands, dual-spin rates.

5.4 Angular motion and guidance signals

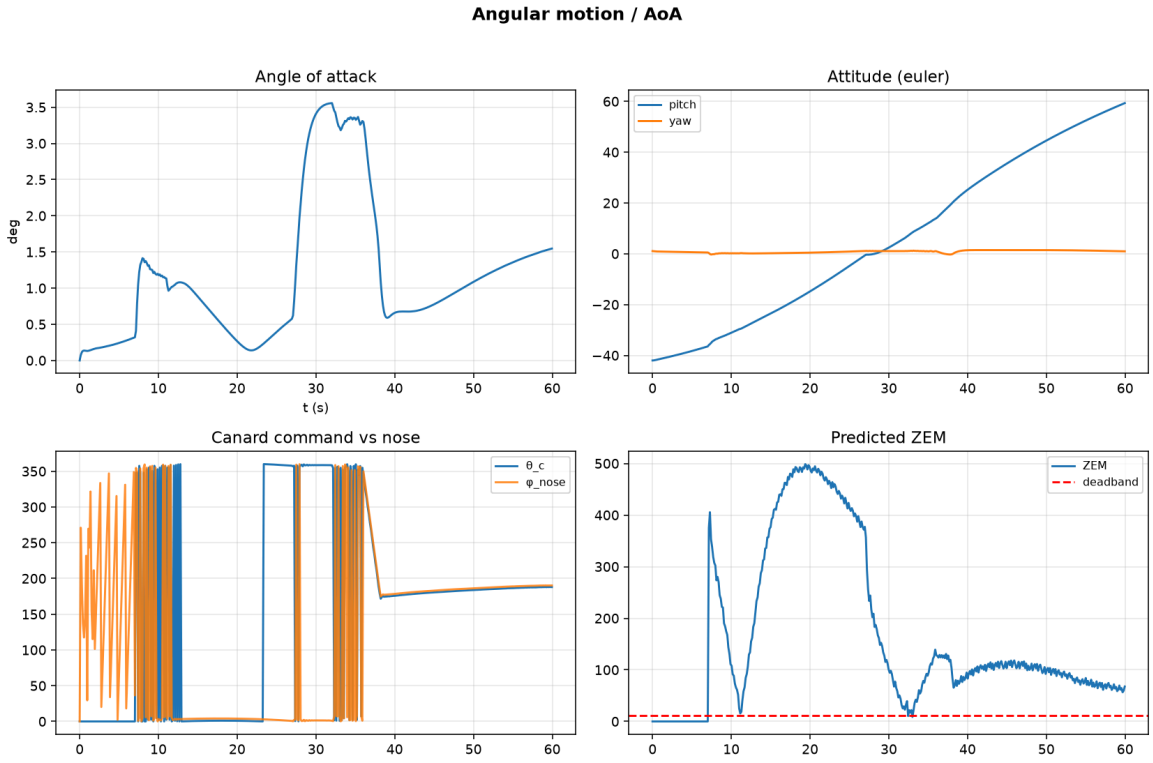


Figure 3. Angle of attack, Euler attitude, canard command vs nose attitude, predicted ZEM.

Angle of attack remains a few degrees. During hold, φ_nose tracks θ_c with lag.

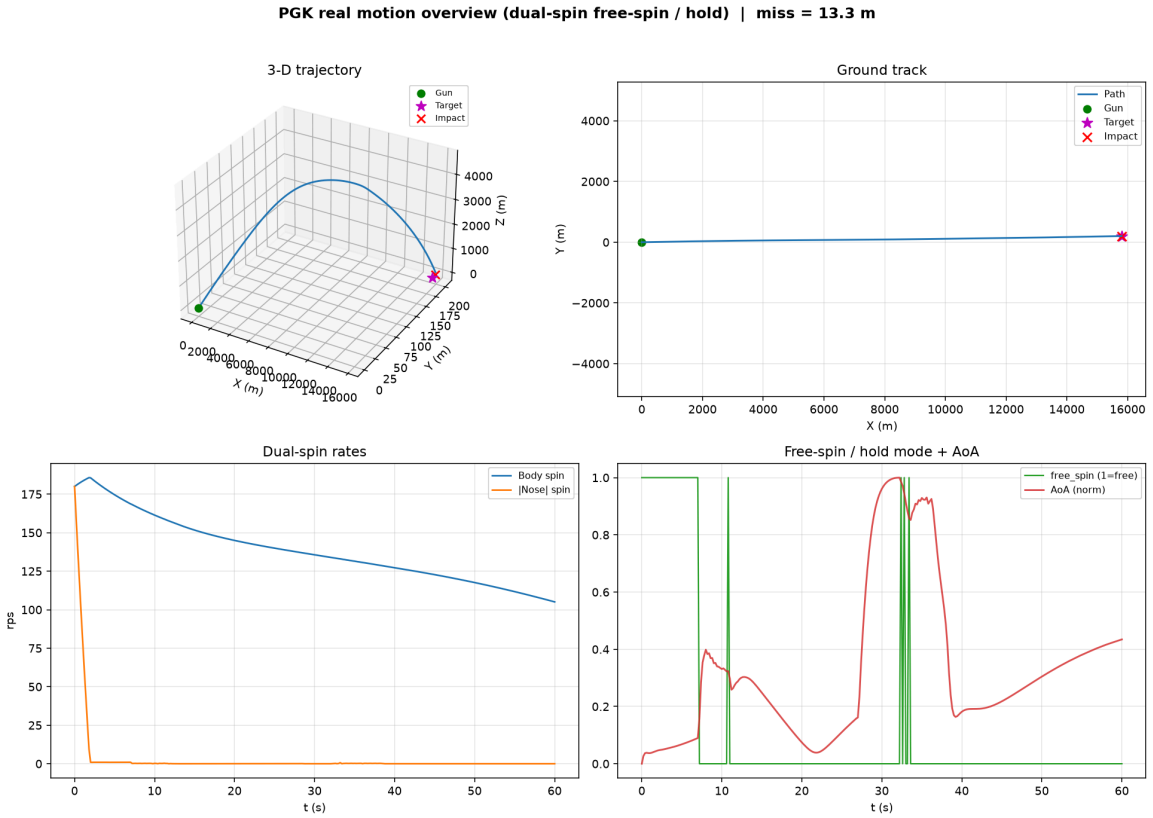


Figure 4. Real-motion overview: trajectory, dual-spin, free-spin/hold and AoA.

5.5 Motion interpretation

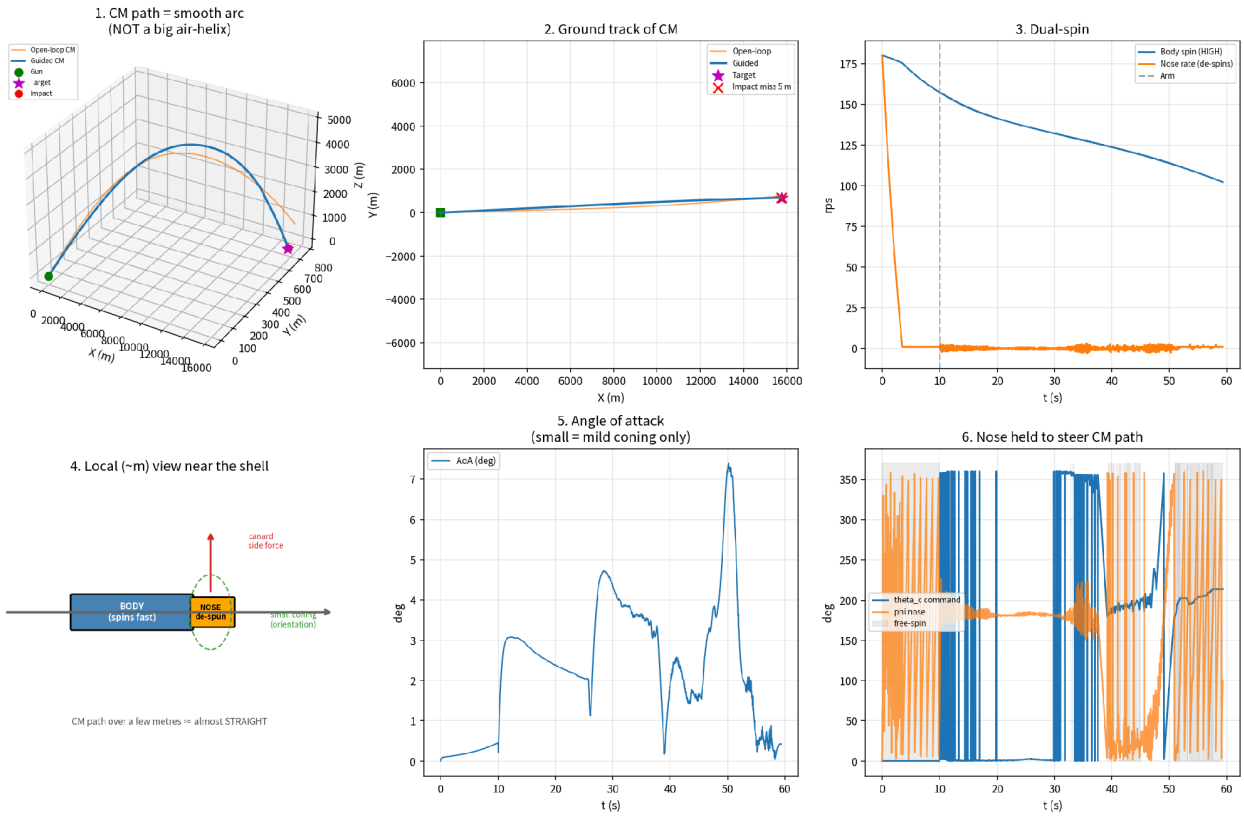


Figure 5. Motion explained: CM path, dual-spin, body/nose schematic, AoA, nose hold.

Over a few metres of path the shell is not helical; spin is about the body axis while the mass centre follows a smooth curve.

5.6 Fuze mode verification

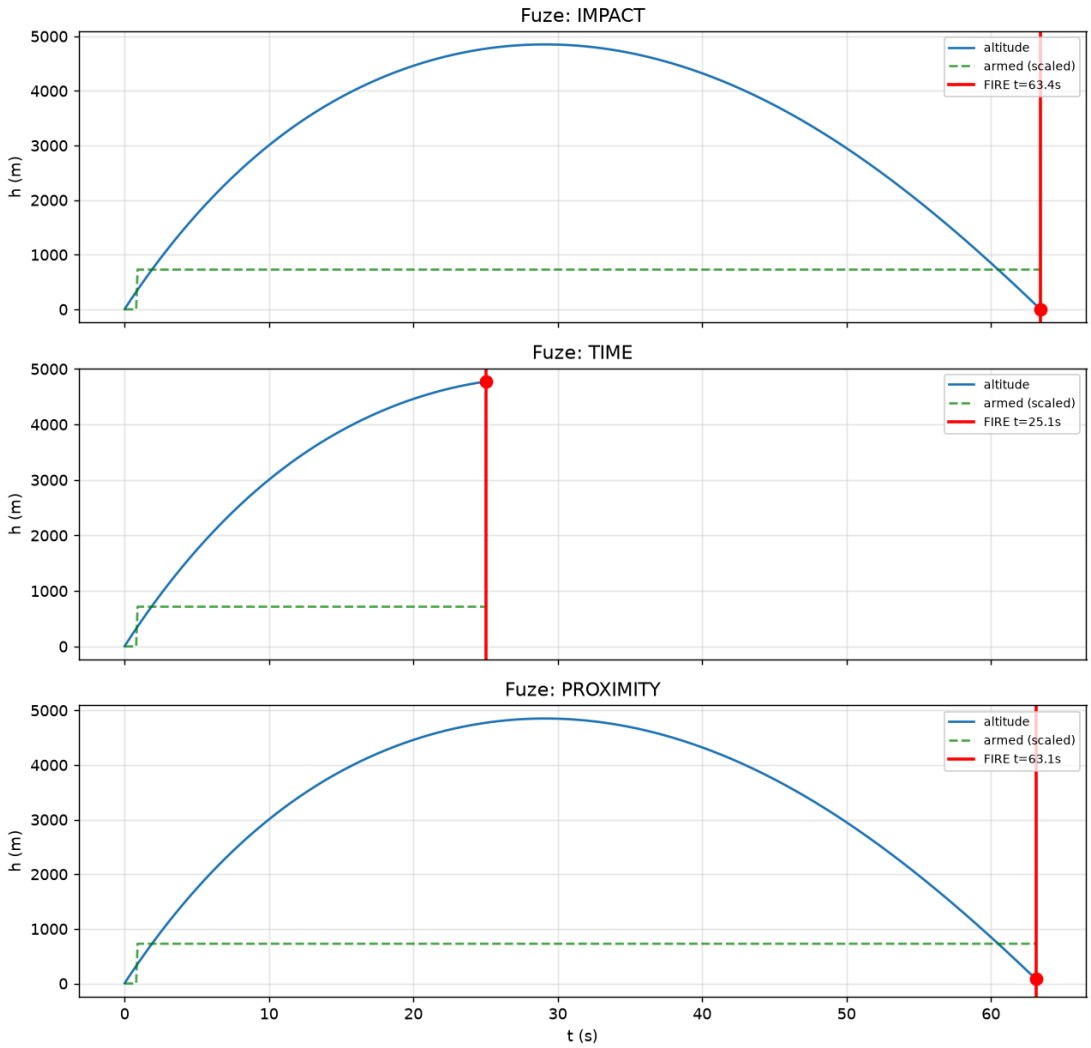


Figure 6. Fuze tests: IMPACT (~63.4 s); TIME (~25.1 s); PROXIMITY near ground.

Safe–arm follows setback. IMPACT and PROXIMITY fire near the end of flight; TIME fires at the programmed epoch.

6. Limitations

Aero tables are engineering-grade, not vehicle-specific CFD. Canard authority is calibrated reduced-order. Single-shot miss is not CEP; Monte Carlo ($N \geq 30$) under dispersions is required for CEP_{50}. High-g electronics and mechanical deployment are not fully co-simulated.

7. Conclusions

A dual-spin fixed-canard PGK GNC model with free-spin/hold control, ZEM guidance, GPS/Kalman navigation, mode FSM, and multi-mode fuzing was implemented. Guided miss of order 13 m improves on 122 m open-loop miss and meets a < 30 m single-shot objective for the reported case. Next steps: formal Monte Carlo CEP and open-loop checks against 155 mm firing-table data.

End of paper.